

SABID BIN HABIB PIAS

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SUMMARY

Results-driven research scientist with expertise in database management, machine learning implementation, and statistical modeling to deliver actionable and responsible insights. Skilled in developing scalable data systems, conducting rigorous experimental studies followed by advanced analyses, and applying Responsible AI principles to ensure fairness, transparency, and human-centered outcomes.

EDUCATION

Indiana University Bloomington

Expected Graduation: December 2025

Ph.D. in Computer Science

· **Research Interest:** Responsible AI, Machine Learning, Natural Language Processing, Data Analysis

Indiana University Bloomington

May 2024

M.S. in Computer Science

Bangladesh University of Engineering and Technology

March 2016

Bachelor of Science in Computer Science & Engineering

PROFESSIONAL EXPERIENCE

Indiana University Privacy Lab

August 2019 - Present

Graduate Research Assistant

Idaho National Laboratory

Summer 2023

Python Programmer Intern

Field Buzz (Dhaka, Bangladesh)

January 2017 - May 2019

Software Engineer

TECHNICAL SKILLS

Languages and Frameworks Python, R, React, PyTorch, PyQt, Flask, LIME, SHAP

ML & LLM Techniques Random Forest, CNN, LSTM, LLM Fine-tuning, LLM Evaluation, RAG

Databases SQL, PostgreSQL, Oracle, NoSQL

Data Analyses Regression, ANOVA, Mixed Effect Model, Factor Analysis, Clustering

Other GPT API, Git, LangChain, Amazon Polly, Google Cloud Platform (GCP)

PUBLICATIONS

- **Sabid Bin Habib Pias**, Ran Huang, Donald Williamson, Minjeong Kim, and Apu Kapadia. The Impact of Perceived Tone, Age, and Gender on Voice Assistant Persuasiveness in the Context of Product Recommendations. In *ACM Conference on Conversational User Interface, CUI 2024 (Best Paper)*
- **Sabid Bin Habib Pias**, Imtiaz Ahmad, Taslima Akter, Adam J. Lee, and Apu Kapadia. Decaying Photos for Enhanced Privacy: User Perceptions Towards Temporal Redactions and ‘Trusted’ Platforms. In *ACM Conference On Computer-Supported Cooperative Work, CSCW 2022*
- Alicia Freel, **Sabid Bin Habib Pias**, Selma Sabanovic, Apu Kapadia: Navigating Trust Erosion in Human-AI Collaboration: Unpacking the Impact of Severity and Timing in Misclassification In *ACM Conference on Fairness, Accountability, and Transparency, FAccT 2025*
- **Sabid Bin Habib Pias**, Alicia Freel, Timothy Trammel, Taslima Akter, Donald Williamson, and Apu Kapadia: The Drawback of Insight: Detailed Explanations Can Reduce Agreement with XAI In *ACM CHI Workshop on Human Centered Explainable AI, HCXAI@CHI 2024*

SELECTED RESEARCH EXPERIENCE

Indiana University Privacy Lab

Mentor: *Dr. Apu Kapadia*

- **Persuasiveness of Conversational Agents' (CA) in Online Shopping:** Evaluated the impact of **CAs' anthropomorphic vocal tones** on consumer engagement through simulated user studies and mixed-methods analysis, achieving an **18% improvement in persuasiveness** with neutral and positive tones over negative ones. Received the **best paper award** at ACM CUI 2024. [PDF]
- **User-Centered XAI:** Conducted usability research on explanation lengths in **explainable AI (XAI)**, discovering a **15% reduction in user acceptance** of lengthy explanations versus no explanation among **users with low technological comfort**. Presented in ACM CHI 2024 workshop on Human-Centered XAI.
- **Privacy Education with LLM:** Generated contextual nudges through **one-shot prompt engineering**, evaluating their effectiveness in guiding users toward privacy-conscious decisions making; achieved a **13% reduction in privacy-violating actions** and a **16% decrease in cognitive load** during task execution.
- **Bitrotting Photos for Enhanced Privacy:** Proposed two temporal redaction methods for enhancing privacy in photo sharing; evaluated the proposed methods in a user study, where 17-21% participants preferred 'non-sensitive' photos. Published and presented the paper at ACM CSCW 2022. [PDF]
- **Decision-Making Awareness in LLM Recommendations:** Designed and prototyped an LLM-powered voice agent using GPT-4 API and Amazon Polly, incorporating linguistic interventions. Conducted semi-structured qualitative studies to evaluate whether these interventions improve user decisions.
- **Assessing Trust Dynamics in Human-AI Collaboration:** Designed a study to investigate the impact of mistake severity and timing on user trust in high-stakes AI systems, suggesting that errors in critical scenarios erode trust regardless of timing, while late errors in less severe contexts are more detrimental. This work has been accepted for a talk at the workshop for Trust and Reliance in Evolving Human-AI Workflows at ACM CHI 2024.
- **Customized Language Learning:** Developed a gamified language learning assistant with conversational correction and reward model by **fine-tuning** a pretrained T5 model using **LoRA**, achieving an **improved BLEU score of 31** through refined tokenization.
- **Decision Awareness in LLM Recommendations:** Designed and prototyped a **voice agent using GPT-4 API and Amazon Polly**, incorporating varying vocal tones. Conducted semi-structured qualitative studies to evaluate whether these interventions improve users' decision-making awareness.
- **Organization Knowledge Assistant:** Developed a RAG-powered enterprise knowledge assistant with GPT-4 and FAISS for improved contextual retrieval, achieving **78% precision and 72% accuracy through semantic chunking and contextual embedding**.

SELECTED INDUSTRY EXPERIENCE

Idaho National Laboratory

Mentor: *Dr. Cody Walker and Dr. Linyu Lin*

- **Explainable AI (XAI) in Power Plant Fault Prediction:** Designed an Explainable AI interface enhancing user interpretability with visual insights; optimized water pump fault prediction using predictive analytics techniques on imbalanced data(Python, PyTorch, LIME, SHAP, PyQT) (**Poster**)

Field Buzz

Mentor: *Habib Ullah Bahar*

- Collaborated with consultants to extract technical requirements; Produced extensive technical reports
- Built data integrity library with an offline data sync mechanism (Android, SQL, PostgreSQL, REST)(**Playstore**)

TEACHING EXPERIENCE

Indiana University

- CSCI-A 542: Technical Foundations of Cybersecurity (Spring 2025)
- CSCI-B 547: Systems & Protocol Security & Information Assurance (Spring 2023)

SELECTED ACADEMIC PROJECTS

- **Language Learning Assistant:** Developed a language learning assistant with grammar correction and conversational practice features by fine-tuning a T5 model using LoRA (Low-Rank Adaptation)
- **Animal Detection from Images:** Designed CNN based architectures and used transfer learning with EfficientDet-D7 for detecting animals from an image subset of ‘Open Image Dataset V6’ (**Notebook**)
- **Music Genre Classification from Waveform Audio:** Designed and compared CNN, RNN and Transfer Learning based architectures for music genre classification on ‘GTZAN’ dataset (**Notebook**)

SCHOLARSHIPS AND AWARDS

- **Best Paper Award** 2024
ACM Conversational User Interface
- **Luddy Research Excellence Award** 2023-24
- **Cognizant Trust and Safety Scholarship** 2023-24
- **Best Undergraduate Database Systems Project, CSE, BUET** 2014

ACADEMIC SERVICES

- **Program Committee (PC) Member** EuroUSEC ’24, ’25
- **Papers Peer Review** CHI’26,’25*, ’24* CSCW’25*, CUI’23*, HRI’25
ICWSM’25, CSCW’24, IMX’24, DIS’24, CUI’24, UIST’24
* *Special recognition for outstanding review*
- **Mentoring Junior Co-chair** SOUPS 2023
- **Undergraduate Research Mentoring** 2024-25, 2023-24, 2022-23
- **Student Volunteer** CHI’24, CSCW’24, CUI’24, CSCW’23, CHI’22, CSCW’22
- **Papers Subcommittee Student Volunteer, Security and Privacy** CHI’22
- **ACM SIGCHI Student Member** 2020-Current

LEADERSHIP EXPERIENCES

- **Luddy Graduate Student Ambassador, Indiana University** 2022- Present
- **Vice President, Bangladesh Student Association, Indiana University** 2021- 2022
- **Graduate Student Representative, Luddy School, Indiana University** 2021- 2022
- **Organising Member, Laboratorian Association of BUET** 2014-15

REFERENCES

- **Dr. Apu Kapadia**
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Luddy School of Informatics, Computing, and Engineering
Indiana University
Email: kapadia@iu.edu
- **Dr. Minjeong Kim**
Professor
Merchandising Department
Eskenazi School of Arts, Architecture and Design
Indiana University
Email: kim2017@iu.edu
- **Dr. Donald Williamson**
Associate Professor
Department of Computer Science and Engineering
Ohio State University
Email: williamson.413@osu.edu
- **Dr. Mary Jean Amon**
Assistant Professor
Department of Informatics
Indiana University
Email: mjamon@iu.edu
- **Dr. Cody Walker**
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Instrumentation, Control and Data Science
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